MAS508-G Control Theory
Lecture 3: State Space Design

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L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – Linear Quadratic Gaussian LQG

Textbook, “Control Systems Engineering”, Chapters 4 &12

L3h. Linear Quadratic Gaussian (LQG) Control - Optimal Control

LQG- Linear Quadratic Gaussian Control

The **linear–quadratic–Gaussian** (**LQG**) **control problem** is one of the most fundamental optimal control problems. It concerns linear systems driven by additive white Gaussian noise.

It is a combination of a linear-quadratic state estimator (LQE) together with a linear-quadratic regulator (LQR). The separation principle states that the state estimator and the state feedback can be designed independently

LTI system

$$\dot{x}(t) = Ax(t) + Bu(t) + v(t)$$
$$y(t) = Cx(t) + w(t)$$

- $\{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ are system, input and output matrices.
- $x(t)$ is states vector and $u(t)$ is input vector.
- $y(t)$ is measurement with noise.
- $v(t)$ is the process disturbance/noise.
- $w(t)$ is the measurement noise.
- $v(t)$ and $w(t)$ are uncorrelated noise, e.g. zero-mean Gaussian white noise stochastic processes.

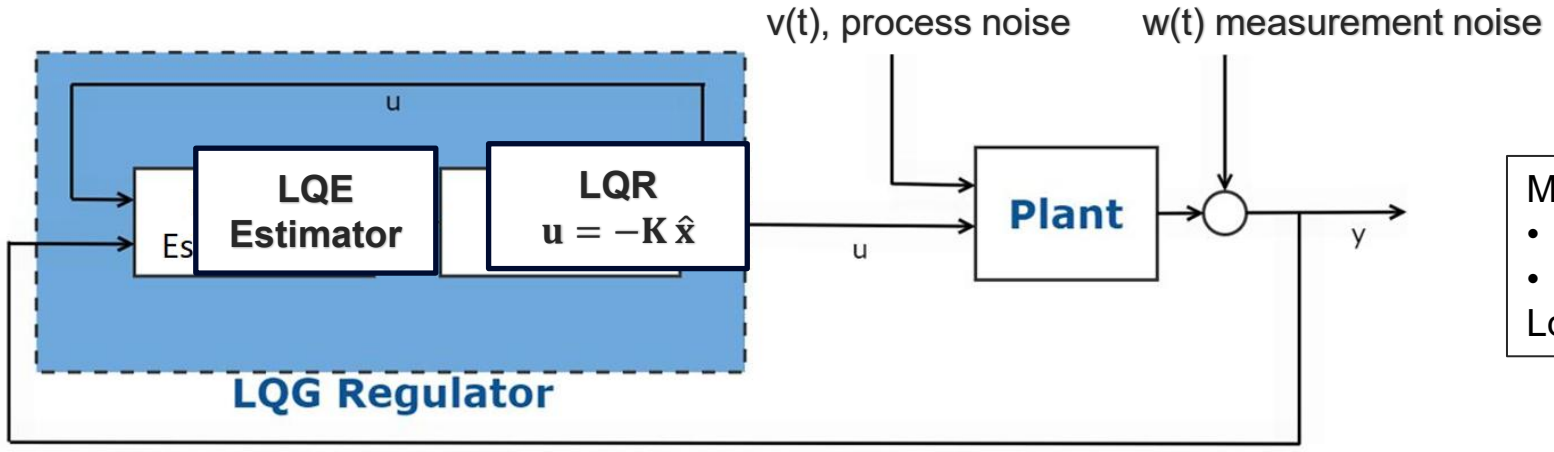
LQG -Output Feedback Control

$$\dot{x}(t) = Ax(t) + Bu(t) + v(t)$$
$$y(t) = Cx(t) + w(t)$$

- It is a combination of a Linear quadratic state estimator (LQE) together with a Linear quadratic regulator (LQR).

LQE Estimator: $\dot{\hat{\boldsymbol{x}}} = \boldsymbol{A}\hat{\boldsymbol{x}} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{L}\,(\boldsymbol{y} - \boldsymbol{C}\hat{\mathbf{x}})$

LQR Control: $\boldsymbol{u} = -\boldsymbol{K}\,\hat{\mathbf{x}}$



Matlab:
- lqgreg(), lqg()
- kalman(), lqr(), lqe()

Lqgreq comine the results of lqr() and kalman()

Linear quadratic state estimation (LQE)

LTI system: $\dot{x}(t) = Ax(t) + Bu(t) + v(t)$

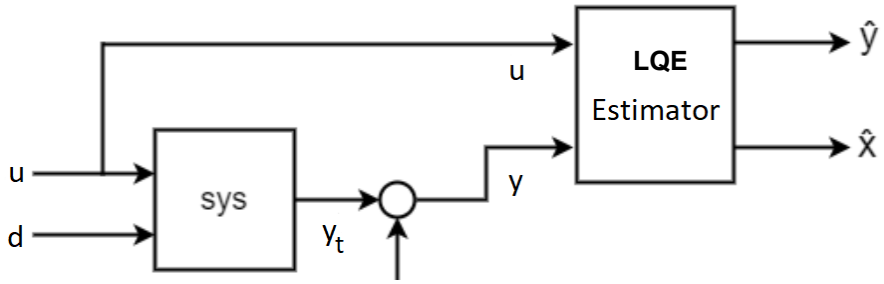
$y(t) = Cx(t) + w(t)$

To minimize the asymptotic norm of estimation error:

$$J = \lim_{t\to\infty} E(\|x - \hat{x}\|^2)$$

LQE:

$$\dot{\hat{x}} = A\hat{x} + Bu + L\,(y - C\hat{x})$$
$$\hat{y} = C\,\hat{x}$$



Disturbance

y_t is the plant true output and does not include the measurement noise.

Kalman filter is a linear quadratic state estimator

Linear Quadratic Estimation (LQE)

$$\dot{\hat{x}} = A\hat{x} + Bu + L\,(y - C\hat{x})$$
$$\hat{y} = C\,\hat{x}$$

To minimize the asymptotic norm of estimation error: $J = \lim_{t\to\infty} E(\|x - \hat{x}\|^2)$

- Observer gain **L**: $L = PC^T W^{-1}$
- P is the solution of Riccati equation:
  $\dot{P} = A^T P + PA + V - PC^T W^{-1} CP$
  $P(0) = E[x(0)x(0)^T]$

W and V are positive definite matrices, associate to the noises $w(t)$ and $v(t)$, such as $E\left(w(t)w(t)^T\right) = W$, $E\left(v(t)v(t)^T\right) = V$, $E\left(v(t)w(t)^T\right) = 0$, $W, V > 0$.

Linear Quadratic Gaussian (LQG) control

LTI system: $\dot{x}(t) = Ax(t) + Bu(t) + v(t),\ y(t) = Cx(t) + w(t)$

LQG Control: $\mathbf{u} = -\mathbf{K}\,\hat{\mathbf{x}}, \qquad \dot{\hat{x}} = A\hat{x} + Bu + L\,(y - C\hat{x})$

To minimize the cost function $\boldsymbol{J}$

$$J = E\left[x^T(T)Fx(T) + \int_0^T x^T(t)Qx(t) + u^T(t)Ru(t)dt\right]$$

$F \geq 0, Q \geq 0, R > 0$, E(.) denote the expected value.

If $T = \infty$, $\boldsymbol{J = \int_0^\infty x^T(t)Qx(t) + u^T(t)Ru(t)dt}$

- Control gain $\mathbf{K}$: $K = R^{-1}B^T S$
- S is the solution of Riccati equation:

$$-\dot{S} = A^T S + SA + Q - SBR^{-1}B^T S, \quad S(T) = F$$

Separation: Estimation + Control

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu} + \mathbf{v}, \mathbf{y} = \mathbf{Cx} + \mathbf{w},$$

Estimator: $\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{Bu} + \mathbf{L}\,(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}}) = (\mathbf{A} - \mathbf{LC})\hat{\mathbf{x}} + \mathbf{Bu} + \mathbf{Ly}$

Control: $\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}}$

- The estimation error $\mathrm{e} = \mathrm{x} - \hat{\mathrm{x}}$

$$\dot{\mathrm{e}} = (\mathrm{A} - \mathrm{LC})\mathrm{e} + \mathrm{LCx} - \mathrm{Ly} + \mathrm{v}$$

$$= (\mathrm{A} - \mathrm{LC})\mathrm{e} + \mathrm{v} - \mathrm{L}\big(\mathrm{Cx} - (\mathrm{Cx} + \mathrm{w})\big)$$

$$\dot{\mathrm{e}} = (\mathrm{A} - \mathrm{LC})\mathrm{e} + \mathrm{v} - \mathrm{Lw}$$

$$\dot{\mathrm{x}} = \mathrm{Ax} - \mathrm{BK}\hat{\mathrm{x}} + \mathrm{v} = (\mathrm{A} - \mathrm{BK})\mathrm{x} + \mathrm{BKe} + \mathrm{v}$$

- Written in matrix as:

$$\begin{bmatrix} \dot{\mathrm{x}} \\ \dot{\mathrm{e}} \end{bmatrix} = \begin{bmatrix} \mathrm{A} - \mathrm{BK} & \mathrm{BK} \\ 0 & \mathrm{A} - \mathrm{LC} \end{bmatrix} \begin{bmatrix} \mathrm{x} \\ \mathrm{e} \end{bmatrix} + \begin{bmatrix} \mathrm{I} & 0 \\ \mathrm{I} & -\mathrm{L} \end{bmatrix} \begin{bmatrix} \mathrm{v} \\ \mathrm{w} \end{bmatrix}$$

- Stability of the closed-loop follows from the triangular structure of this system and the fact that the matrices $A - BK$ and $A - LC$ are both Hurwitz.
- Compute the characteristic polynomial of the closed-loop system

$$det\left(sI - \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix}\right)$$
$$= det\begin{bmatrix} sI - (A - BK) & -BK \\ 0 & sI - (A - LC) \end{bmatrix}$$
$$= det[sI - (A - BK)] \cdot det[sI - (A - LC)]$$

- The two equations above are called the **separation property**, which allows us to **calculate the feedback gain K and the observer matrix L independently** as required if the system $(A, B, C)$is controllable and observable.
- **LQG: K and L are obtained by minimizing estimation error and cost function.**

Example: LQG

Design a LQG to a spring-mass system:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + v, \qquad y = [1 \quad 0]x + w$$

where covariances of v and w are I_2 and 0.1, respectively.

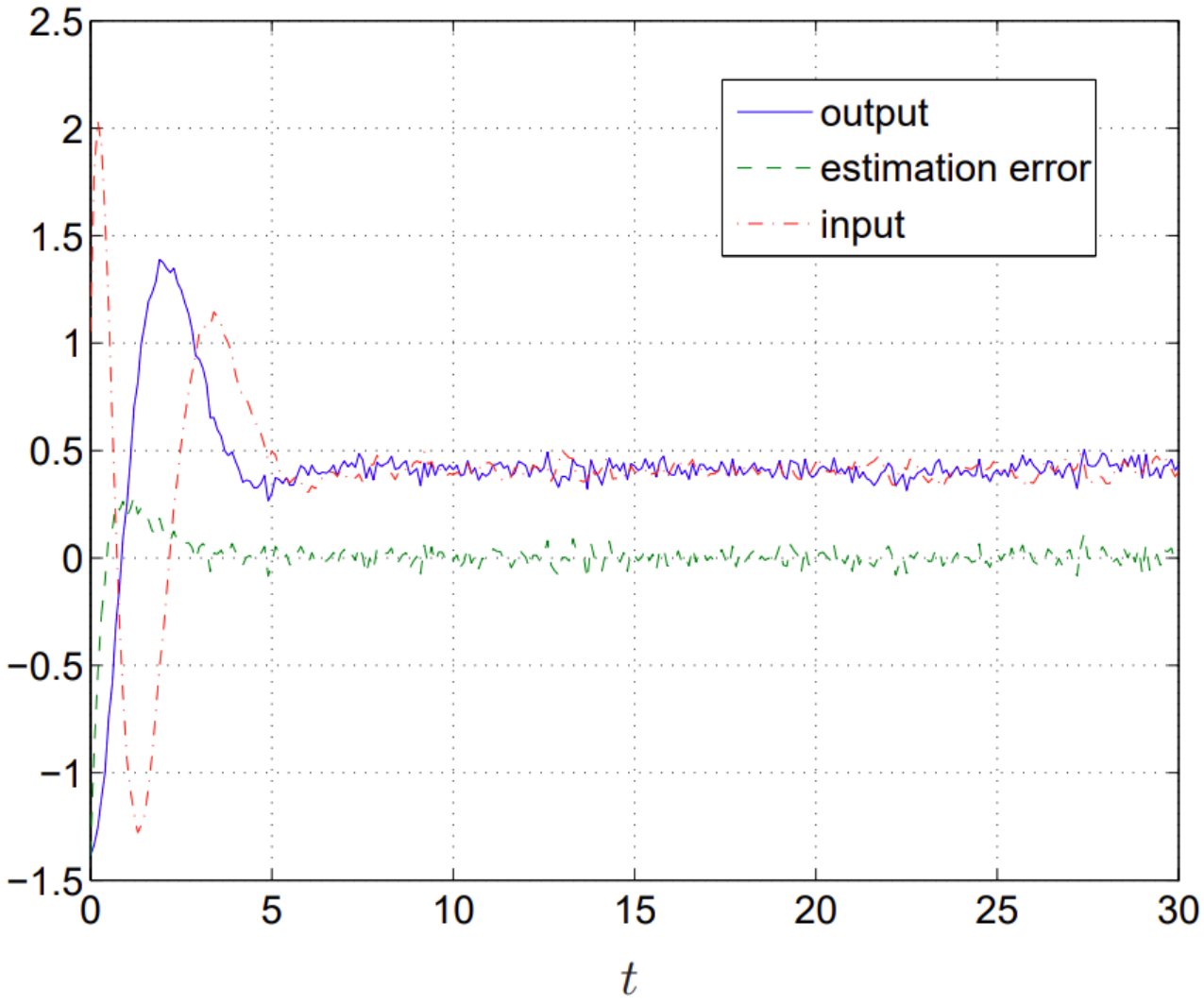
LQG:

- Check system $(A, C) observable, and\ (A, B)\ controllable$.
- Choose the following parameters:

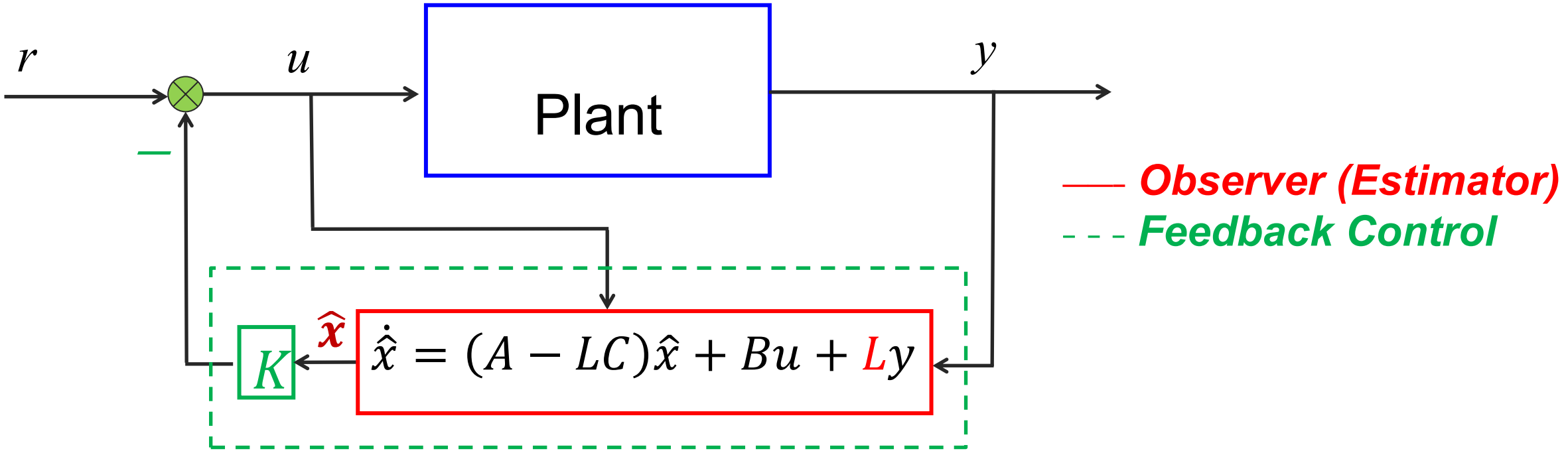
$$Q = 5C^T \times C, \qquad R = 1, \qquad V = I_2, \qquad and \quad W = 0.1.$$

MATLAB codes

```
n = 2; A = [0 1;-1 0]; B = [0 1]'; C = [1 0];
Q = 5*C'*C; R = 1;
K = lqr(A,B,Q,R);
V = eye(n); W = 0.1;
L = lqe(A,eye(n),C,V,W);
```


output
estimation error
input
2.5
2
1.5
1
0.5
0
−0.5
−1
−1.5
0
5
10
15
20
25
30
t

Summarize: Output feedback control



2 Methods to find control gain and observer gain

1) Pole placement method to calculate K and L
2) Optimal method to calculate K and L (Linear quadratic method)

Exercise

Inverted Pendulum: State-Space Modelling and Control Design

State-Space Modelling

https://ctms.engin.umich.edu/CTMS/index.php?example=InvertedPendulum&section=SystemModeling

Control Design

https://ctms.engin.umich.edu/CTMS/index.php?example=InvertedPendulum&section=ControlStateSpace